Advanced topics in regression

control regressors, colinearity, and statistical power

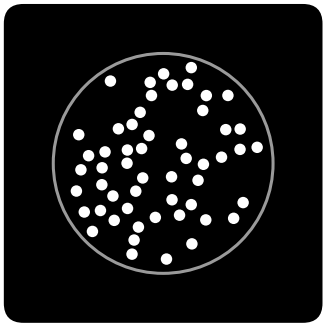
Alexandre Hyafil

Data Analysis for Cognitive Neuroscience



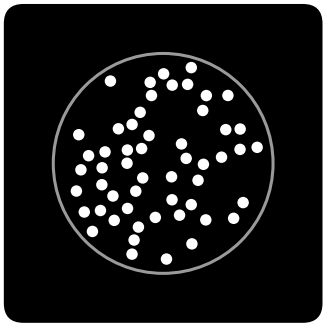


Looking for numerosity in the brain



“more or less than 50 dots?”

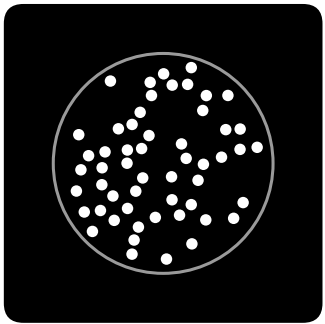
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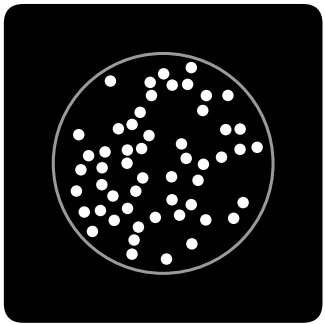
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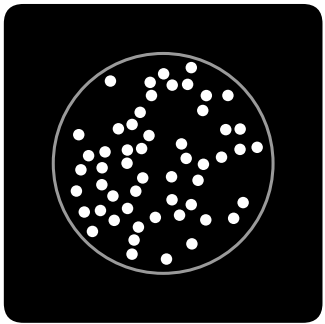
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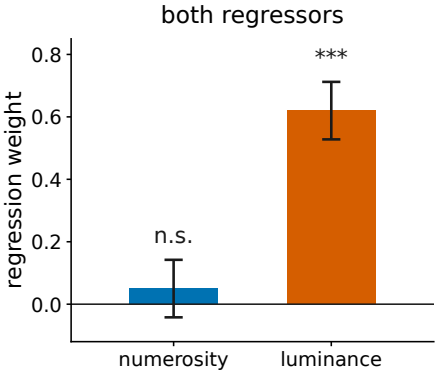


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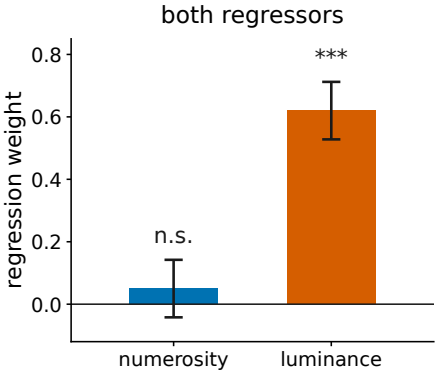
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$$\text{activity} \sim w_0 + \underbrace{w_1\,\text{numerosity}}_{\text{what we are after}} + \underbrace{w_2\,\text{luminance}}_{\text{what we must control}}$$

The same data, two different conclusions

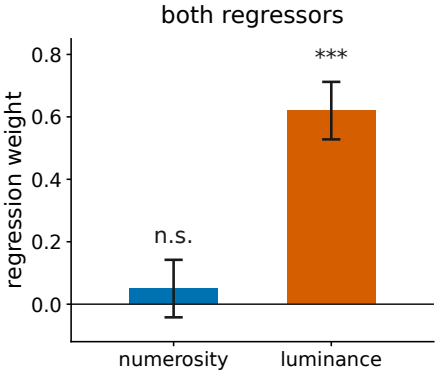


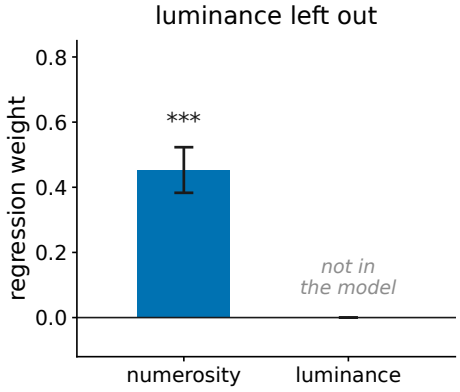
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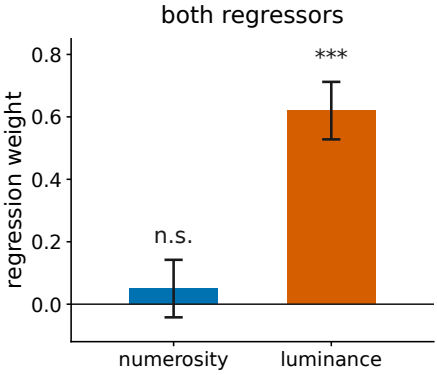
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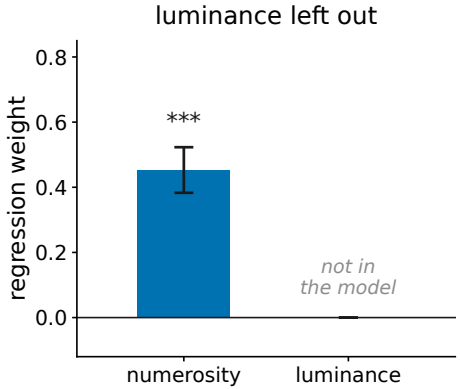


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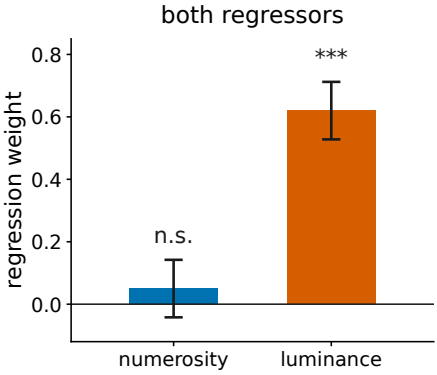


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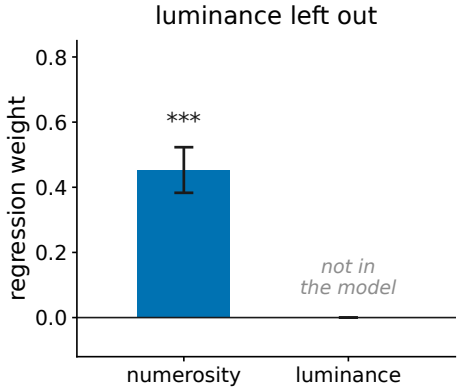


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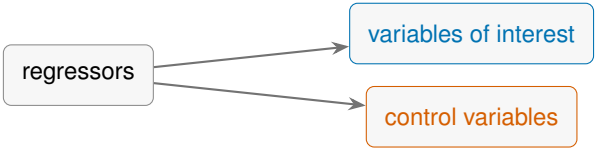
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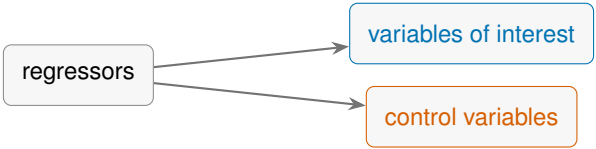
An omitted regressor does not disappear: its effect is absorbed by whatever correlates with it.

Variables of interest, and control variables



(“nuisance regressors”)

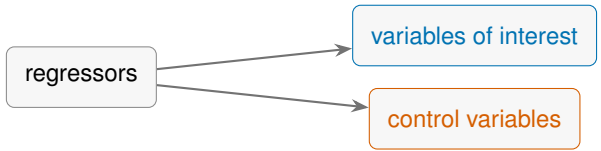
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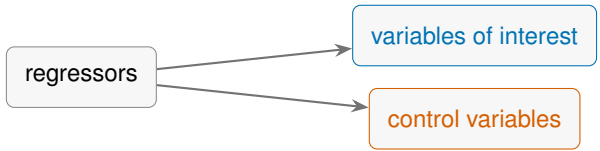


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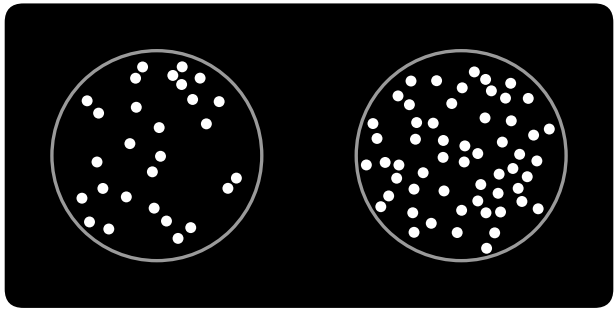


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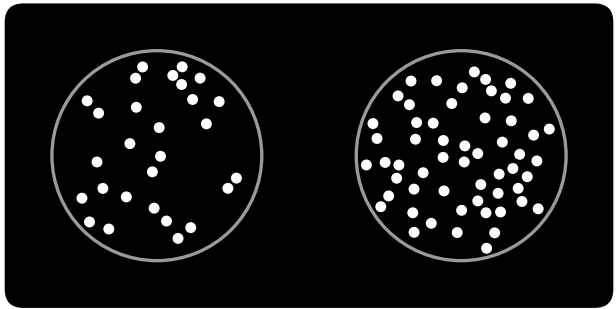
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- **Not for free**, when it is not: every extra regressor costs precision, and with a weak effect and few observations that cost is **overfitting**, hence a poorer estimate

The issue with colinear regressors



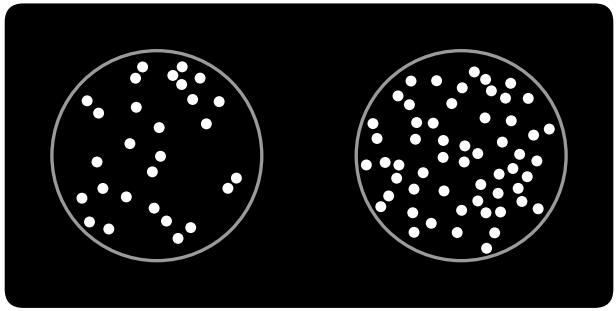
N = 28 N = 56

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- the two regressors are **proportional**: no dataset, however large, can tell their contributions apart

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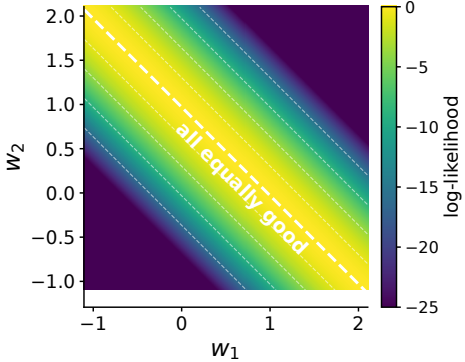
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the maximum is a whole **line**: the model is **unidentifiable**

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- so the categories *together with the intercept* are colinear
- the remedy: drop one category — the **reference level**, against which the others are read — or drop the intercept

Reading the weights of a categorical regressor

orientation estimation on seven consecutive days — *does the bias drift across the week?*

the day enters as a **categorical** regressor, `Mon` as reference

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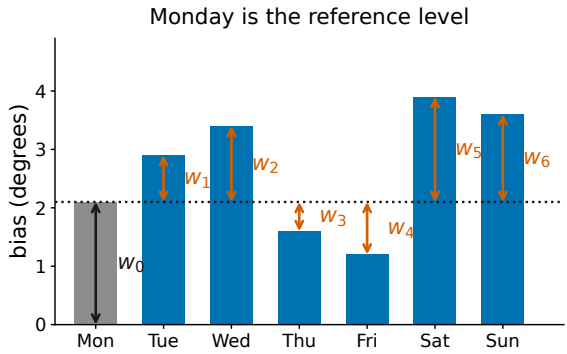
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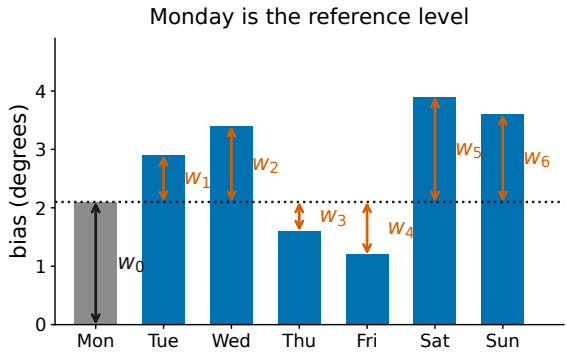
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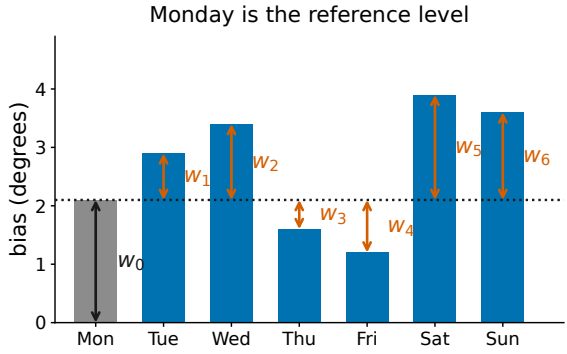
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- change the reference, and all of them change meaning



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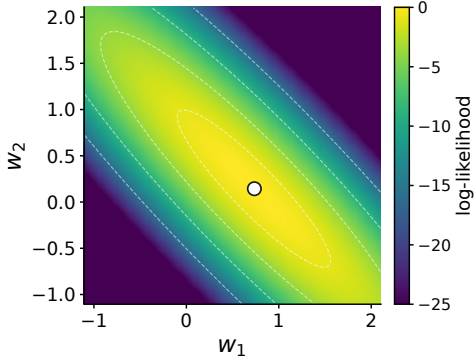
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one maximum again — but the likelihood is nearly **flat** along one direction

Detecting it in advance: the Variance Inflation Factor

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- a rule of thumb, not a law: above 5, think again
- but $\mathrm{SE}_{\text{ideal}}$ itself shrinks as $1/\sqrt{n}$: **collecting more data** is the other way out of a high VIF — the one thing that never rescues *colinear* regressors

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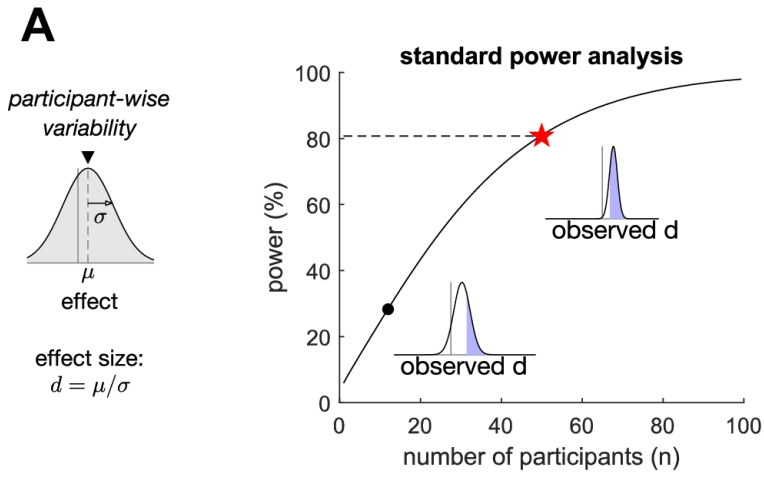
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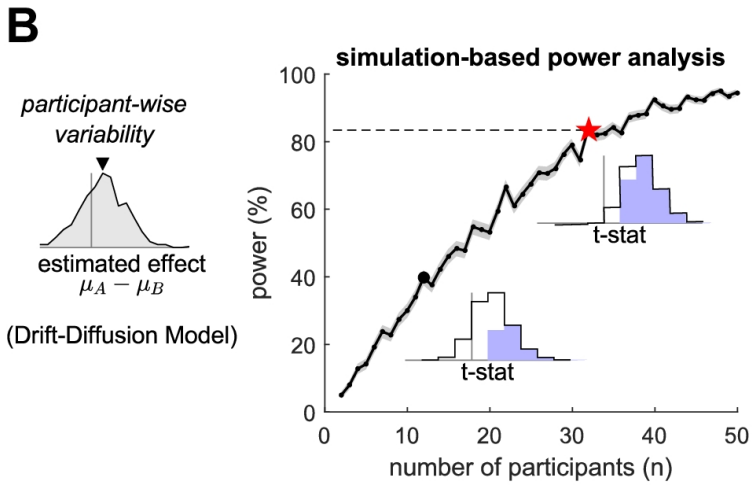
- the complement of a **miss**: $1 - \beta$
- it grows with the effect size, with the sample size, and with cleaner measurements
- we aim for 80–90% — and it has to be worked out **before** collecting the data

Power analysis with a formula



Barbosa, Stein, Zorowitz, Niv, Summerfield, Soto-Faraco & Hyafil (2023), *Behavior Research Methods* 55:58–76 (CC BY 4.0), Fig. 2A

Power analysis by simulation



Barbosa, Stein, Zorowitz, Niv, Summerfield, Soto-Faraco & Hyafil (2023), *Behavior Research Methods* 55:58–76 (CC BY 4.0), Fig. 2B